MECH 325 - Midterm Book 1
Closed Book

Tuesday, October 26, 2022

Solution

Name: _____

Student Number: _____

Circle your section

Section 102 (11:00- 12:30)

Section 101 (9:30 – 11:00)

Signature: _____

Part 1 MC Mark _____ / 12

Part 1 SA Mark _____ / 10

Part 2 LA Mark _____ / 38

Total _____ / 60

Instructions

There are two parts to this exam with different instructions for each. Please read carefully.

Part 1 – Closed-Book (22 marks)

Multiple choice (12 marks: 1 marks per question). Complete all 12 questions by marking your response on the paper.

Short answer (12 marks). Complete all 5 questions by marking your response in this exam booklet.

After you have completed Part 1, hand in your paper in order to begin the Open Book portion. You may not return to Part 1 after you hand it in.

Part 2 – Open-Book (38 marks)

Long answer (38 marks): Complete all parts to the questions for Part 2 by marking your responses on the exam paper. This portion of the exam is open-book and open-notes. Laptops and tablets are permitted but must be disabled for communication and access to the internet.

NO COMMUNICATION DEVICES!

Part 1A: Multiple Choice (12 Points)

Answer all 12 questions. Each is worth 1 mark. Choose the single best response – a given question may have more than one choice that is correct but marks will only be given for the *best* answer.

1. **What is the best description of the type of motion between the teeth of two properly meshed spur gears?**

   a) Purely rolling
   b) Purely sliding
   c) Some rolling when teeth engage and disengage, but predominantly sliding
   d) Some sliding when teeth engage and disengage, but predominantly rolling
   e) Approximately equal parts sliding and rolling

2. **For spur and helical gears, the pressure line is tangent to which circles?**

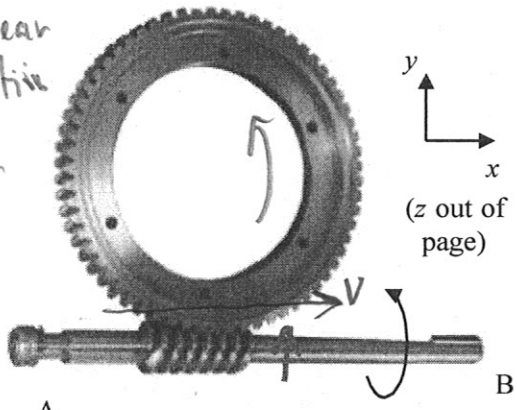
   a) Pitch Circle
   b) Base Circle
   c) Addendum Circle
   d) Dedendum Circle
   e) Clearance Circle

3. **The worm gear shown has a 1" pitch diameter and has two threads. The gear has an 8" pitch diameter and 96 teeth. If the worm turns 24 revolutions in the direction indicated, how far does the gear turn?**

   a) 8 rotations clockwise
   b) 4 rotation clockwise
   c) ½ rotation clockwise
   d) ½ rotation counter-clockwise
   e) 4 rotations counter-clockwise

Dual thread means advancing two gear teeth per revolution
24 revs = 48 teeth
= ½ rotation

answer →



Note: Will also accept (c) as direction is not clear on figure.

4. **For a well-designed gear set (helical or spur) transmitting a torque form a pinion to a driven gear, the radial loading on the shaft is due predominantly to:**

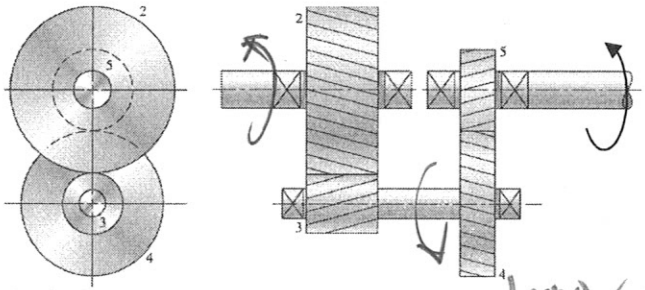
   a) Diametral pitch
   b) Involute profile
   c) Helix angle
   d) Pressure angle
   e) Face width

4. **A spur gear has 40 teeth, a 14° pressure angle, and a pitch diameter of 2.0 in. What is the diametral pitch of the gear?**

40 teeth $P_D = N/d_p = 40/2 = 20$

a) 0.05
b) 1.42
c) 2.86
d) 7.0
e) 20.0

6. For the gear train shown, gear 5 is the driving pinion Gear 5 has 17 teeth, Gear 4 has 51 teeth. Gear 3 has 18 teeth and Gear 2 has 36 teeth. An 1800 rpm motor is driving the gear box in the Clockwise direction. What is the speed and direction of the output shaft?



$\frac{17}{51} \cdot \frac{18}{36} = \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{6}$

a. 300 rpm clockwise
b. 300 rpm counter clockwise
c. 360 rpm clockwise
d. 360 rpm counter clockwise
e. 600 rpm clockwise

$1800/6 = 300$

same direction

7. The motor attached to the gearset in Question 4 above is rated at 1000 Watt. What is the torque on the output shaft?

Power $= T\omega = 1000 = T(300\text{rpm} \cdot 1\text{m}/60\text{sec}) \cdot 2\pi$

$T = 1000/5\text{rps} \cdot 2\pi =$

31.8

a) 13.3 N-m
b) 19.9 N-m
c) [redacted] N-m 31.8
d) 37.7 N-m
e) 53.1 N-m

8. The term *"Pre-Extreme Horespower"* refers to the following:

a) The breaking strength of a synchronous belt
b) The power rating for a chain where roller fatique or excessive wear are the modes of failure
c) The power rating for a chain where link fatigue failure is the primary mode of failure
d) The maximum horsepower rating for a chain
e) The smallest degree of lubrication

9. A V-belt drive has a driving pulley operating at 200 rpm, a drivetrain value of e = 0.25, a nominal power of H_{nom} = .75 hp, a service factor of K_s = 2.5, and a design factor of n_d = 1.5. TWO belts are to be used in parallel. The following belts are available in the catalogue. Which is the smallest belt that will satisfy the design requirements?

e = 0.25

$H_{design} = (H_{nom}) K_s \cdot n_d$

$= (0.75)(2.5)(1.5) = 2.81$ HP

Two belts @ 1.5 HP = 3.0 HP > 2.81 HP

a) Allowable power H_a = 0.7 hp
b) Allowable power H_a = 1.1 hp
c) Allowable power H_a = 1.5 hp
d) Allowable power H_a = 2.9 hp
e) Allowable power H_a = 5.7 hp

10. Which of the following statements is *always true* for a planetary gear train with multiple planet gears? (assume the planetary gear train is not connected to anything else)

a) The planetary gear train has more degrees of freedom than a compound gear train T
b) All of the planetary gears must be of different diameters from each other F
c) Only one of the planetary gears may mesh with the sun gear F
d) There must always be an even number of planetary gears F
e) The planetary gears can never provide a reverse rotation F

11. Rank the following flexible drives from lowest to highest power capacity. Assume approximately equal dimensions and operating conditions.

a.	V-Belt	Flat belt	Roller Chain
b.	V-Belt	Roller Chain	Flat belt
c.	Flat belt	V-Belt	Roller Chain
d.	Flat belt	Roller Chain	V-Belt
e.	Roller Chain	Flat belt	V-Belt

12. A reasonable safety factor when specifying wire rope for a high-speed passenger elevator is:

a. 2
b. 2 to 2.5
c. 3 to 4
d. 9 to 11
e. 20 to 30

13 (Bonus) To convert from ft-lbs/sec to horsepower use the following conversion

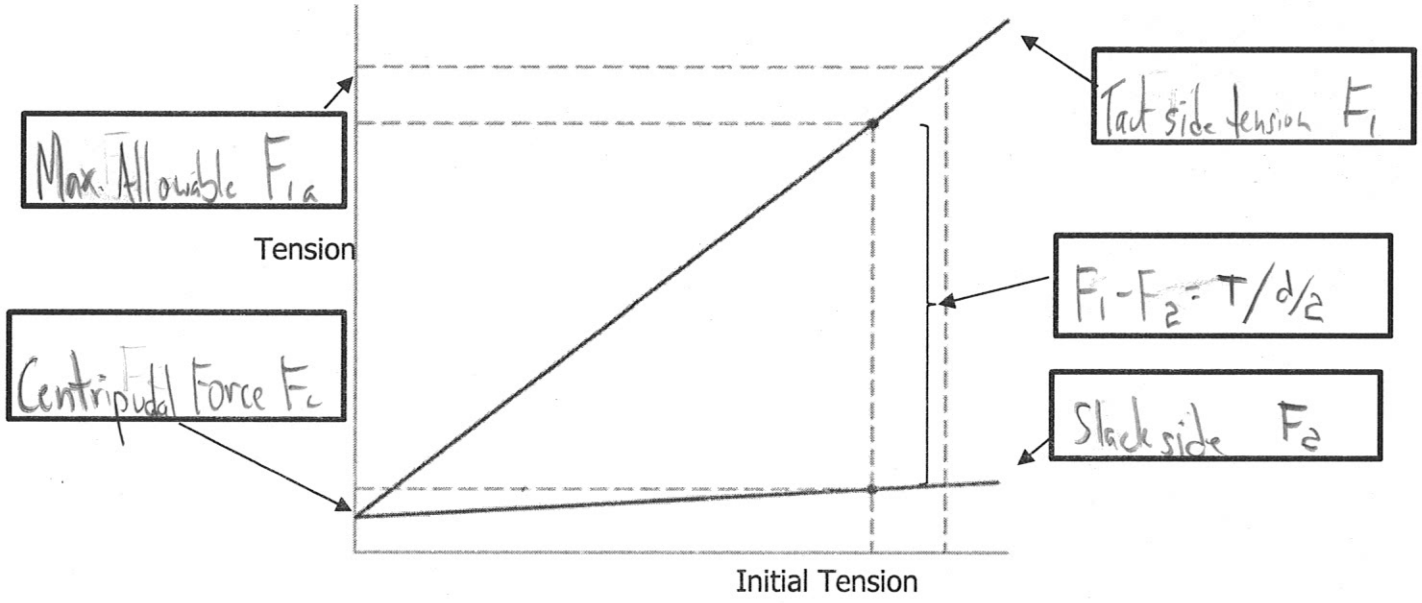
550 ft·lb/sec = 1 HP

a) Multiply by 454
b) Divide by 33000
c) Multiply by 550
d) Divide by 550
e) Divide by 32.2

Part 1B – Short-Answer Questions (10 marks)

Answer all questions in the spaces provided.

1. The relationship for flat belt tension versus initial tension is shown in the diagram below. Complete the diagram by giving the name or equation for each of the five quantities identified. Put your answers in the boxes provided. (3 marks)



2. Describe how the quantities above are used to determine the power in the belt drive system. What is meant by "effective friction?" (3 marks)?

Power in the drive belt system is determined by the difference between the taut (tight) side and slack side tensions. Power = Tω with torque determined as $T = (F_1 - F_2)/d/2$. Increasing F_i will provide higher friction force which will increase the delivered power.
The effective friction (friction development) f' is a derived value that is based only on belt tension ratio and wrap angle. If $f' < f$ we know the drive will deliver the power

16. In gear design, what are the advantages of helical gears over spur gears? Are there any disadvantages? (2 marks)?

Advantages
- higher transmitted force for same pitch diameter
- smoother delivery of torque
- quieter operation (less vibration)

Dis-advantage
- greater an axial force component
- higher cost

21. a) Why is lubrication important in a chain drive train system? (1 mark)
b) What is meant by "galling?" (1 mark)

Lubrication is important as it:
- reduces friction in moving part (links, rollers)
- significantly extends life of chain
- improves mechanical efficiency
- protects part from oxidation
- reduces heat generation

Galling is a form of adhesive wear. Under very high pressure some metals will fuse together and tear when subsequently moved. Aluminium stainless steel & titanium are especially prone to galling

MECH 325 - Midterm Book 2
Open Book

Tuesday, October 26, 2022

Name: _____

Student Number: _____

Circle your section

Section 101 (11:00-12:30)

Section 102 (9:30-11:00)

I adhered to the policy of No Communication Devices.

Signature: _____

Instructions

Open notes and texts are allowed. Laptops are allowed but all communication features must be disabled.

No Communication Devices Allowed!

There are **TWO** Problems to complete.

Part 2 – Open-Book (38 pts)

Long answer: Complete all parts to the question by marking your response in the pages provided. This portion of the exam is open-book and open-notes. **At the end of the exam, return this handout with your answers in the booklet.**

This is a multi-part problem. For each part, use the parameter values given. If you cannot complete one part, you may assume some values and move on in order to receive marks for subsequent parts. Clearly indicate your final answer to each part by drawing a box around it. Separate different parts with a line drawn across the page.

Long Answer Problem (38 marks)

Problem 1 (14 Marks)

A synchronous (timing) belt system is required to attach an electric motor to a centrifugal pump. The following information relates to the equipment

Driver	AC Motor (Normal Torque)
Driven Machine	Pump (Centrifugal)
Service	18 hours/day
Input Speed	1160 rpm
Input Power	6 HP
Output Speed (Nominal)	600 rpm
Minimum Centerline Distance	18 inches

Select a suitable synchronous belt system for the application. Specify the belt size and sprocket sizes. Is the belt speed suitable? What is the safety factor for this system?

(2mk) Velocity ratio $VR = 1160/600 = 1.933 \approx 2.000$

Service Factor SF Driver = AC Motor (NT) Driven: Pump (cent.)

From Table 7-8 S.F. = 1.7 (Continuous Service 16-24 hrs/day

(1mk) Design Power = S.F (Power input) = 1.7(6) = 10.2 HP

(1mk) From Figure 7-27 select 8mm G7 style belt (10.2 H.P @ 1160 rpm)

For velocity ratio of 2.000 candidate belts include

Table 7-7

DriveR	DriveN
22	44
24	48
28	56
32	64
36	72

Table 7-9 Rated Power 30mm belts

22 teeth @ 1160 rpm $H_{rated} = 8.06 < 10.2$ HP

24 " " " $= 9.37 < 10.2$ HP

28 " " " $= 12.0 > 10.2$ HP

32 " " " $= 14.5 > 10.2$ HP

36 $= 17.1 > 10.2$ HP

(2mk)

Best candidate is 28 & 56 tooth sprockets

Next CD > 18.0 inches Table 7-7 gives the following belt

(1mk) Select 1280 8MGT 160 Tooth belt. Pitch length = 50.394 inch

CD = 18.53"

Belt correction Factor

Table 7-11 $C_L = 1.10$ for 1280-8MGT 160 teeth

2 mk $H_{adjusted}$ = Base Rated Power * length correction factor, C_L

$H_{adjusted} = (12.0)(1.1) = 13.2$ HP

2 mk Safety Factor $n_{sf} = \frac{H_{rated}}{H_d} = \frac{13.2}{10.2} = 1.3$

Table 7-4

1 mk

	Sprocket #
28 tooth sprocket Pitch dia. = 2.807"	P28-8MGT-30
56 tooth sprocket Pitch dia. = 5.614"	P56-8MGT-30

Belt: 1280 8MGT 160 tooth belt P.L. = 50.394"

Not Required: Bushing Specifications: belt 28 tooth sprocket Bushing 1108

28 tooth sprocket 56 tooth sprocket Bushing 2012

Belt Speed

2 mk $V_{belt} = \omega r = \frac{PD_{driving}}{2} n_{driving} = \frac{2.807}{2} \frac{1160 rpm}{12 in/ft} \cdot 2\pi$

$V_{belt} = 852.4$ ft/min < 3500 fpm Acceptable

Note: If 50 mm belt is selected or we get

1280-8MGT-50

$H_{adjusted} = (20.8 \text{ HP})(1.1) = 22.9$ HP ← Table 7-10

$n_{SF} = 22.9/10.2 = 2.2$ very healthy s.f.

If 24:48 sprockets are chosen $n_{sf} = 1.01$

Problem 2 (24 Points)

Part 1 of 3

(9 points) Instead of using the synchronous belt, it is proposed that a gear box be used to drive a higher capacity pump (**input power = 15HP**) at a required **nominal** speed of 600 rpm. Do an analysis on the following full depth involute **HELICAL** gears:

Note: you can choose imperial or metric units

Driver	AC Motor (Normal Torque)
Driven Machine	Pump (Centrifugal)
Service	18 hours/day
Input Speed	1160 rpm
Output Speed	600 rpm (nominal)
Input Power	15 HP (10 kW)
Pressure Angle (Normal)	20°
Helix Angle	15°
Diametrical Pitch (Transverse)	12 teeth/inch (module = 2.0 mm.)
Pinion	20 teeth
Gear	? teeth

Determine the following:

a) Given 20 teeth for the pinion, select a gear and determine the gear ratio (velocity ratio)
b) The pitch diameter for the pinion and gear (transverse only)
c) Pitch line speed
d) Tangential, radial, and axial forces for the gears

a) velocity ratio = 1160/600 = 1.933

of gear teeth = (1.933)(20) = 38.66 Round up to 39

1 mk

Hence gear ratio = 39/20 = 1.95 (only 1% difference)

b) Pitch dia = No. of teeth / Diametrical Pitch

2 mk

d_{pinion} = 20/12 = 1.666 inches

d_{gear} = 39/12 = 3.250 inches

Transverse

(Eq'n 8-16)

Normal Diametrical Pitch $P_{nd} = P_d / \cos\psi$ $\psi = 15°$

Not required

d_{np} pinion = 1.667/cos ψ = 1.726 inches

d_{np} gear = 3.250/cos ψ = 3.365 inches

c) Pitch line speed

Pitch line speed = tangential velocity $= \omega r$

(1mk)

$$V_t = \frac{\pi D n}{12} \text{ ft/min} \quad \text{(Eq'n 8-35)}$$

$$= \frac{\pi (1.667)\, 1160}{12} = \boxed{506.2 \text{ fpm} = V_t}$$

d) Tangential, Radial, & Axial Force

i) Power = torque × rotational speed

$T = \text{power}/\omega = 63000\, P/n$ [lb.in] [Eq'n 10-4]

(2mk)

Power = 15 HP $n = 1160$ rpm

$T = 63000(15)/1160 = 814.7$ lb-in

W_t = Torque/pitch radius $= T/d/2 = 814.7/1.667/2$

$W_t = 975.6$ lbs

ii) Radial Force $W_r = W_t \tan \phi_t$ [Eq'n 10-8]

(2mk)

$\tan \phi_n = \tan \phi_t \cos \psi$ [Eq'n 10-1]

$\tan \phi_t = \tan \phi_n / \cos \psi$

$\phi_t = \tan^{-1}[\tan 20^\circ / \cos 15^\circ] = 20.65^\circ$

$\therefore W_r = (975.6 \times \tan 20.65^\circ) = 367.7$ lbs

iii) Axial Force $W_x = W_t \tan \psi$

(1mk)

$W_x = (975.6 \times \tan 15^\circ) = 261.4$ lbs

$W_t = 975.6$ lbs
$W_r = 367.7$ lbs
$W_a = 261.4$ lbs

Part 2 of 3

(9 Points) Determine the AGMA **bending** stress number for the 20 tooth **pinion ONLY**.

Given:	
Face Width	2.00 inch (50 mm.)
Pressure Angle (Normal)	20°
Diametrical Pitch (Transverse)	12 teeth/inch (module = 2.0 mm.)
Gear Quality	$A_v = 8$ ($Q = 9$)
Material for Pinion & Gear	Steel
Gear Box Enclosure	Commercial Quality
Size Factor	$K_s = ?$
Rim thickness factor	$K_B = ?$
Overload Factor	$K_o = 1.50$ (uniform power source and moderate shock)
Dynamic factor	$K_v = ?$
Load Distribution factor	$K_m = ?$ (For Shigley Text: $C_{mc} = C_{pm} = C_e = 1.0$)
Geometry factor	J= ?
Bending stress number	$s_t = ?$

AGMA Bending Stress Eq'n

$$s_t = \frac{W_t P_d}{F J} K_o K_v K_m K_B K_v$$

[M.H 9-27, Sh.sty 14-15]

1) Size Factor $\boxed{K_s = 1.0}$ since $P_d > 5$ (1 mk)

2) Rim Thickness Factor $\boxed{K_B = 1.0}$ since pinion dia = 1.667 in (1 mk)

3) Overload Factor $\boxed{K_o = 1.50}$ (Given) (1 mk)

4) Dynamic Factor K_v

Pitch line velocity $V_t = 506$ ft/min ≈ 500 fpm

Gear quality $A_v = 8$

$B = 0.25[A_v - 5]^{0.667} = 0.52$

$C = 50 + 56(1.0 - B) = 76.87$

$K_v = \left(\frac{C}{C + \sqrt{V_t}}\right)^{-B} = \boxed{1.14 = K_v}$

$\boxed{K_v = 1.14}$ [Figure 9-16] (1 mk)

5) Load Distribution Factor K_m

pinion proportion factor C_{pf}

$$C_{pf} = \frac{F}{10 D_p} - 0.0375 + 0.0125F$$

$$C_{pf} = \frac{2}{10(1.667)} - 0.0375 + 0.0125(2$$

$C_{pf} = 0.107$

$F/D_p = 2.0/1.667 = 1.2$

From t. 9-18 $C_{pf} \approx 0.10$ @ $F = 2''$, $F_m = 1.2$

Mesh alignment factor C_{ma}

$C_{ma} = 0.127 + 0.0158(F) - 1.093\times10^{-7}F^2$

$C_{ma} = 0.127 + 0.0158(2) - 1.093\times10^{-7}(2)^2$

$C_{ma} = 0.158$ [0.16 from Fig 9-13]

(2mk)

$K_m = 1.0 + 0.107 + 0.158 = 1.27$

$\boxed{K_m = 1.27}$ $K_m = 1.26$ using figure1

Geometry Factor J

For Helical Gear use Fig. 10-6 to determine J

Fig 10-6 a) $J = 0.46$ @ $\psi = 15°$ & $N_p = 20$ teeth

(2mk) Fig 10-6 b) Mult. factor $K = 0.95$ @ $\psi = 15°$ & $N_G = 39$ teeth ≈40 teeth

$\therefore$ Modified $J = (0.46)(0.95) = 0.44$

$\boxed{J = 0.44}$

Bending Stress Number $s_t = \dfrac{W_t P_d}{F J} K_o K_v K_m K_B K_v$

with W_t = 975.6 lbs $P_d = 12$ $F = 2.00$ we get

(1mk) $s_t = \dfrac{(975.6)(12)}{(2)(0.44)}(1.50)(1.14)(1.27)(1.0)(1.0)$

$\boxed{s_t = 29090 \text{ lbs/in}^2 \simeq 29 \text{ ksi}}$

Note: Shigley has $J' = JK = (0.51)(0.93) = 0.47$

so $s_T \simeq 27$ ksi

↑ Figure 14-7 ↖ Fig. 14-8

Part 3 of 3

(6 points) Select a suitable through hardened steel for the **pinion**. Specify the alloy and heat treatment.

The gear is expected to operate for 18 hours a day for 200days/ year for 4 years. Define your expected reliability and temperature factors.

What is your safety factor for the pinion bending stress?

Comment on the selection of the proposed helical gear. Is this analysis complete?

We require an acceptable material strength number that give a safety factor of approx. $1.0 < n_{sf} < 1.5$

(1 pt)

$$S_{at}' = S_{at} \frac{Y_n}{(SF) K_R (K_T)}$$ ← Shigley

Assume $K_R = 1.0$ for 99% reliability

$K_T = 1.0$ at $T < 120°C$

Estimate Stress Cycle Factor Y_n

$$\#\text{ cyles} = (1160\text{ rpm})(60\text{ min/hr})(18\text{ hr/day})(200\text{ day/yr})(4\text{ years})$$

load cycles $N_c = 1.0 \times 10^9$

(2 pt)

From Fig. 9-21 $Y_N = 0.95$

$Y_N = 1.3558 (1.0 \times 10^9)^{-.0178}$

$Y_N = 0.94$ eqn. in Fig. 9-21

with $S.F = 1.0$

$$S_{at}' = (29\text{ ksi})(0.95)/(1.0)(1.0)(1.0) = 27.5\text{ ksi}$$

From Fig 9-18 We are on the low end of the through hardened gears. For $S_{at} = 27.5$ ksi $HB \approx 195$

(1 pt)

For Grade 1 steel $S_{at} = 0.0773\,HB + 12.8$ ksi

$$27.5 = 0.0773\,HB + 12.8$$

$$HB_{min} = 191$$

Several materials are available such as 4140 OQT 1300 HB = 235

$S_{at} = 0.0773(235) + 12.8 = 31.0$ ksi

1mk

safety factor: $\frac{S_{at}}{S_t} = \frac{31.0 \text{ ksi}}{27.5 \text{ ksi}} = 1.13$

Comment:

According to AGMA, safety factor of 1.13 is reasonable if $k n_{Fr} < 1.5$

However, our experience has shown that n_{Fr} for contact stress is usually lower than n_{Fr} for bending stress.

1mk I might expect this selection to fail in contact stress and we will need to alter the design by i) increase face width, ii) higher quality gear materials iii) increase gear sizes. We need to perform contact stress analysis to confirm our assumption.

Also, we need to determine the axial pitch to ensure we have a minimum of two teeth engagment.

$$F/p_x > 2.0$$